PRAGYA PANT

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EDUCATION

Master of Science in Geographic Information Science
Minnesota State University, Mankato | *Mankato, Minnesota*
GPA: 4.00 **Anticipated: May 2027**

Bachelor of Engineering in Geomatics Engineering
Kathmandu University | *Nepal*
CGPA: 3.51 **Graduated: Jun 2022**

EXPERIENCE

Student Worker Paraprofessional Senior | *Minnesota Department of Transportation | St. Paul, Minnesota*
- Working on Blowing Snow Control project creating maps for snow fence and snow traps.**Aug 2026 - Present**
- Analyzing blowing snow data for creating safer roads during extreme winter across Minnesota.

Graduate Teaching Assistant | *Minnesota State University, Mankato | Minnesota* **Aug 2025 - Present**
- Supporting students in GIS, cartography, spatial analysis, and ArcGIS Pro workflows, for coursework.
- Conducting lab activities for spatial data handling, map creation, and natural resources management.

GIS Intern | Austin Utilities | *Austin, Minnesota* **Jun 2026 – Aug 2026**
- Carried out mapping and analysis of spatial data in utility management projects.
- Supported in planning field surveys using ArcGIS Online and ArcGIS Enterprise within the engineering department.

Teaching Assistant | *Kathmandu University | Nepal* **Dec 2022 - Apr 2025**
- Supervised practical coursework on GIS, GPS, remote sensing, surveying, and cartography.
- Supported field training sessions on topographic, cadastral, alignment, and hydropower survey.

Data Quality Intern | *Humanitarian OpenStreetMap Team (HOT)* **Oct 2022 - Dec 2022**
- Validation of data quality in OpenStreetMap collected from field surveys and open mapping.
- Hands-on experience on open-source field data with and shapefile, GeoJSON, and XML file formats.

GIS Project Manager | *Open Mapping Hub- Asia Pacific* **Apr 2022 - Jun 2022**
- Managed team and project workflow for rooftop farming feasibility research.
- Ensured quality control of open-source collected data.

PUBLICATIONS

Monitoring Water Turbidity in the Chiang Rai Reach of the Mekong River Using Sentinel-2 NDTI
Rajamangala University of Technology Lanna (RMUTL) Engineering Journal **Nov 2025**
https://doi.org/10.14456/rmutlengj.2025.14

Sustainable land use planning in developing countries using GIS and multi-criteria analysis: a case study of Lalitpur district, Nepal
City and Built Environment **Jul 2025**
https://doi.org/10.1007/s44213-025-00050-x

Land Cover Change Before and After the Sendai Framework of Disaster Risk Reduction
Social Science Research Network **Sep 2024**
http://dx.doi.org/10.2139/ssrn.5012977

SKILLS

GIS & Mapping: ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, QGIS, Map Layouts, Cartography
Data Visualization and Modeling: Web maps, Spatial data management
Data Formats: Shapefiles, Geodatabases, Vector data, Raster data, CSV, KML, GeoJSON
Remote Sensing: Image Interpretation, Land use/Land cover Analysis, Disaster Planning
Software: ArcGIS Pro, QGIS, ENVI, AutoCAD, Microsoft Office, TransCAD, TerrSet, ENVI
Soft Skills: Analytical Thinking, Technical Communication, Time Management, Collaboration
Programming Skills: C, C++, Java, Python, R

PROJECTS

- Remote Sensing and Machine Learning Approach to Mapping Potential Permafrost Zones in the Himalaya Region (Ongoing Master's Thesis)
- Assessing Urban Growth and Vegetation Change in Phoenix City, Arizona 2015-2025
- Modeling and Simulation of Urban Micro-Climate Variations in Minnesota State University, Mankato
- Monitoring, Modeling, and Hazard Assessment of Glacial Lake Outburst Flood (GLOF) in Nepal
- 3D Modelling using photogrammetry and Feasibility of UAV and Total Station in Cadastral Mapping
- Field Survey for Hydropower, Route, Survey using Total Station, UAV, and topographic map preparation using AutoCAD and GIS software

CONFERENCE AND SEMINAR PRESENTATIONS

Speaker: "Open- Source Mapping for Climate Change and Disaster Resilience"
5th Young Surveyors Network Asia and the Pacific Meeting, Nepal **Nov 2024**
Guest Speaker: "Empowering Geospatial Careers"
International Day of Girl Child **Oct 2024**
Paper Presentation: Pant, Pragya, Narayan Thapa, Sujan Nepali, and Suman Sanjel, Land Cover Change Before and After the Sendai Framework of Disaster Risk Reduction | FIG Regional Conference 2024 Kathmandu, Nepal **Sep 2024**
Conference Presentation: "The OSM Spectrum: Versatility of OSM Across Different Users and Communities" State of the Map Conference, Nairobi, Kenya **Sep 2024**

ADDITIONAL TRAINING

Social Sensing and Big Data Computing for Disaster Management | Wuhan University, China **Jul 2024**
LIESMARS (State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing)

AWARDS, GRANTS, AND OTHER SYNERGISTIC ACTIVITIES

- Dr. Mary T. Dooley Endowed Geography Field Studies Award | *MNSU, Mankato* **Apr 2026**
- James F. Goff Endowed Geography Graduate Research Award | *MNSU, Mankato* **Apr 2026**
- Graduate Teaching Assistantship | *MNSU, Mankato* **Aug 2025**
- FIG Foundation Grant | *FIG Foundation* **Sep 2024**
- International GeoInformatics Summer School Scholarship | *LIESMARS, Wuhan University, China* **May 2024**
- HOT x DataCamp Donates Scholarship | *Humanitarian OpenStreetMap Team* **Mar 2024**
- Travel Grant | *State of the Map (SoTM) 2024* **Feb 2024**
- Climate Action at Home Grant | *Youth Sounding Board-Nepal, EU Delegation to Nepal* **Jul 2023**

PROFESSIONAL AFFILIATION

Gamma Theta Upsilon (GTU) | Member, International Geographic Honor Society **Apr 2026**